

The RVNA Protocol: A Systems Architecture for Post-Exposure Clearance of Rabies Virus in the Central Nervous System (Version 2.1)

A spatiotemporal biophysical transport framework combining convection-enhanced parenchymal delivery, multi-target virucidal neutralization, intracellular polymerase arrest, metabolic assembly deprivation, and gated neuroplastic reconstruction

The RVNA Collaborative

Correspondence: rvna.protocol@proton.me | Permanent Identifier: DOI: 10.5281/zenodo.22675931

Executive Summary | The near-total fatality of symptomatic rabies encephalomyelitis exposes a failure of physical macromolecular transport rather than an unalterable biological constraint. Following peripheral inoculation, virions ascend axoplasm via dynein-dependent retrograde motors and establish replication behind the blood-brain barrier (BBB). At this stage, systemic biologics fail to achieve virucidal concentrations within deep subcortical and brainstem nuclei. Historical metabolic coma protocols, including the Milwaukee regimen, failed across international trials because bioenergetic suppression does not clear parenchymal viral reservoirs or halt trans-synaptic circuit destruction. Here, we delineate the biophysical transport framework for the RVNA Protocol (Version 2.1), an investigational multi-modal systems architecture engineered for post-exposure central clearance. The platform replaces concentration-driven diffusion with stereotactically guided, positive-pressure convection-enhanced delivery (CED), establishing bulk interstitial advection across the thalamus, dorsal pons, and medulla oblongata while integrating dedicated cervical subarachnoid micro-perfusion to preserve phrenic motor pools (C3–C5). Virucidal neutralization is spatially stratified: full-length monoclonal IgG1 (SYN023) clears the bulk interstitial bath, while compact single-domain VHH nanobodies (Rab-E8/H7) partition into the narrow 20 to 30 nm synaptic cleft to arrest cell-to-cell spread. Concurrently, synthetic adenosine analogues (galidesivir) induce non-obligate RNA-dependent RNA polymerase chain termination, 2'-MOE antisense gapmers direct catalytic RNase H1 cleavage of viral mRNAs, and atorvastatin deprives viral assembly of host lipid droplet platforms. Structural neuroplastic repair via exogenous neurotrophin micro-infusion (BDNF, NT-3, NGF) is strictly restrained behind an assay-confirmed virological clearance gate ($\Gamma(t) = 1$), preventing premature cytoskeletal polymerization from accelerating viral dissemination. This framework delineates the transport kinetics, biophysical models, formulation parameters, and emergency regulatory pathways (21 CFR Part 312) required for clinical salvage in symptomatic lyssavirus encephalitis.

1 The Post-Exposure Gap: Cold Chains, Administration Errors, and Clinical Mortality

Rabies virus causes an acute, progressive encephalomyelitis with a case-fatality rate approaching 100% once clinical signs emerge [1]. The World Health Organization (WHO) classifies human rabies as a neglected tropical disease, with an estimated 95% of global mortalities concentrated across Africa and Asia and approximately 40% occurring in pediatric populations under 15 years of age [2, 3]. Standard passive surveillance metrics frequently under-ascertain this burden. Quantitative epidemiological modeling of solid-organ donor transmission clusters suggests that as few as 7% of domestic rabies fatalities in the United States between 2000 and 2024 were formally confirmed in national surveillance registries, demonstrating substantial under-detection even within a high-capacity healthcare system [4]. While direct surveillance data from low- and middle-income countries reflect distinct epidemiological dynamics, the U.S. under-ascertainment signal raises a broader concern: the canonical global estimate of 59,000 annual fatalities may be biased downward wherever diagnostic and death-registration systems systematically miss encephalitic rabies.

Prompt, protocol-adherent post-exposure prophylaxis (PEP) confers near-total protection against viral progression when initiated without delay [2, 5]. In resource-limited settings, decentralized delivery networks routinely encounter devastating lo-

gistical failures: cold-chain breaches, unpowered depot refrigeration, volatile procurement schedules, and chronic inventory stockouts [2, 5, 6]. Provider administration errors introduce additional therapeutic failure modes. Inactivated cell-culture rabies vaccines require validated intradermal or intramuscular delivery into the deltoid or anterolateral thigh. Gluteal administration remains strictly contraindicated due to unpredictable adipose deposition and attenuated antigenic presentation, both of which impair seroconversion kinetics [7, 8]. For severe Category III exposures involving transdermal bites or scratches, the immediate infiltration of rabies immunoglobulin (RIG) directly into and around all wound margins is mandatory to provide localized passive neutralization during the induction of active endogenous immunity.

Omitting RIG creates an open neural conduit. Active vaccine antigens require 7 to 14 days to elicit circulating virus-neutralizing antibody (VNA) titers exceeding the correlative threshold of 0.5 IU/mL [5]. In the absence of immediate local passive neutralization, virions bind low-affinity neurotrophin receptors ($p75^{NTR}$), neural cell adhesion molecules (NCAM), and nicotinic acetylcholine receptors (nAChR) displayed across the neuromuscular junction [9, 10, 11]. Retrograde dynein motor complexes haul the viral ribonucleoprotein (RNP) centripetally along peripheral axoplasm directly into the central axis [9].

Central neuroinvasion upends therapeutic pharmacology.

Once inside the axis, the pathogen is physically sequestered behind the blood-brain barrier (BBB), rendering standard post-exposure measures useless. Inpatient clinical care remains entirely palliative: physicians manage autonomic storms, hydrophobia, and respiratory arrest without altering the clinical outcome. Experimental interventional regimens have repeatedly failed. Most visibly, the Milwaukee Protocol, which utilized drug-induced metabolic coma via ketamine and midazolam alongside supplemental antivirals, failed to demonstrate reproducible clinical efficacy [12]. Subsequent international evaluations confirmed only isolated, non-reproducible survivors alongside at least 64 documented protocol failures, prompting explicit recommendations to abandon drug-induced metabolic suppression in favor of direct neuro-pharmacological interventions targeted at underlying neuropathology [13, 14, 15, 16, 17]. The RVNA architecture addresses this therapeutic void by formulating symptomatic rabies as a spatiotemporal biophysical transport and clearance problem requiring direct intracerebral delivery, sustained viral replication suppression, and localized immunomodulation.

2 Pharmacokinetic and Biophysical Constraints on CNS Interventions

2.1 Pharmacological Limitations of Systemic Metabolic Suppression

Parenchymal penetration is fundamentally restricted by the hydrodynamic and microstructural properties of the brain extracellular matrix. Inducing metabolic coma reflects a flawed conception of central nervous system kinetics during fulminant viral encephalitis. Alan C. Jackson articulated the clinical consensus:

“There is no scientific rationale for the use of therapeutic coma, and there are numerous reports of failures using this approach. Therapeutic coma should not be used for the therapy of rabies.” [13, 17]

Metabolic coma rests on an untenable premise. Proponents hypothesized that depressing cerebral metabolic oxygen consumption ($CMRO_2$) would buy sufficient time for host adaptive immunity to clear the pathogen. This assumption collapses against lyssavirus biology. Microvascular exclusion, rapid trans-synaptic circuit propagation, and active intracellular immune subversion preclude spontaneous viral elimination. Systemic immunoglobulins and small-molecule antivirals fail to achieve protective concentrations within deep subcortical and brainstem targets.

2.2 Intra-Axonal Sequestration and Innate Immune Evasion

Pathogen persistence within the host nervous system is driven by distinct molecular evasion mechanisms:

- Retrograde intra-axonal transit physically sequesters the viral ribonucleoprotein core inside endosomal compartments, preventing direct exposure to circulating neutralizing antibodies during transit from peripheral nociceptors to dorsal root ganglia and the spinal cord [9].
- Following soma invasion, the viral phosphoprotein (P protein) orchestrates targeted intracellular immune evasion. P

binds phosphorylated STAT1 and STAT2 complexes, preventing karyopherin-dependent nuclear translocation and suppressing downstream transcription of Type I interferons ($IFN-\alpha/\beta$) [6, 18, 19].

- Concurrently, P inhibits TANK-binding kinase 1 (TBK1) signaling complexes, arresting the liquid-liquid phase separation required for host antiviral stress granule condensation while driving Negri body viral factory condensation, preserving an uninhibited cytoplasmic replication niche [6, 20].

2.3 Endothelial Permeability and the Blood-Brain Barrier Mass Sink

Under physiological conditions, continuous tight-junction complexes among brain microvascular endothelial cells (principally claudin-5, occludin, and zonula occludens-1) restrict passive paracellular transit across the neurovascular unit [21, 22]. The classical 500 Da heuristic describes a broad permeability trend for passive paracellular transport, but BBB penetration depends on molecular size, charge, lipophilicity, hydrogen bonding, transporters, efflux mechanisms, and molecular architecture. Regardless of the exact permeability threshold, full-length antibodies do not achieve therapeutically reliable parenchymal exposure across the intact BBB at clinically practical systemic concentrations [22].

Neuroinflammation provides no therapeutic opening. While late-stage lyssavirus encephalitis triggers focal inflammatory cascades that alter endothelial integrity, this barrier breakdown occurs heterogeneously and unpredictably across the neuroaxis. High-dose systemic infusion of therapeutic immunoglobulins results in extensive peripheral elimination and off-target tissue binding in the liver, spleen, and kidneys. This peripheral mass sink consumes the circulating dose, while interstitial concentrations across infected subcortical nuclei remain sub-therapeutic.

2.4 Mass Transport Dynamics: Interstitial Diffusion Constraints

Direct delivery of biologics into the cerebrospinal fluid (CSF) via intrathecal or intracerebroventricular routes fails to bypass parenchymal transport limits. The brain parenchyma functions as a dense, fluid-saturated porous medium in which macromolecular transport without an external pressure gradient is governed strictly by passive Fickian diffusion. The characteristic diffusion timescale (t) over an effective transport distance (L) is described by:

$$t \approx \frac{L^2}{2D^*} \quad (1)$$

where D^* represents the effective tissue diffusion coefficient, related to the aqueous diffusion coefficient (D_0) through tissue tortuosity (λ) and extracellular volume fraction (α):

$$D^* = \frac{D_0}{\lambda^2} \quad (2)$$

For the RVNA reduced-order transport calculation, an illustrative effective diffusion coefficient of $D_{IgG}^* \approx 1.0 \times 10^{-7} \text{ cm}^2/\text{s}$ is used for a 150 kDa IgG in grey matter ($\lambda \approx 1.6$), based on the extracellular-space diffusion framework [23, 24].

The kinetic consequence predicted by diffusion modeling is severe:

- Interstitial diffusion across an inter-capillary distance of $100\ \mu\text{m} = 0.01\ \text{cm}$ requires approximately 8.3 minutes:

$$t = \frac{(0.01\ \text{cm})^2}{2(1.0 \times 10^{-7}\ \text{cm}^2/\text{s})} = 500\ \text{s} \approx 8.3\ \text{minutes} \quad (3)$$

- Transit across a 1.0 mm cortical layer requires approximately 13.9 hours.
- Over a distance of 1.0 cm (the scale required to reach deep subcortical and brainstem autonomic centers from ventricular surfaces), passive diffusion requires approximately 57.8 days:

$$t = \frac{(1.0\ \text{cm})^2}{2(1.0 \times 10^{-7}\ \text{cm}^2/\text{s})} = 5.0 \times 10^6\ \text{s} \approx 57.8\ \text{days} \quad (4)$$

Because acute symptomatic rabies results in irreversible autonomic collapse within 3 to 10 days of clinical onset, passive diffusion from the ventricles or subarachnoid space creates a direct transport bottleneck, requiring timeframes that vastly outrun the clinical survival window.

2.5 Convection-Enhanced Delivery Mechanics

Convection-Enhanced Delivery (CED) resolves these kinetic limitations by replacing concentration-driven diffusion with positive-pressure bulk interstitial flow [25, 26, 27]. Stereotactically positioned, reflux-resistant micro-cannulas infused via precision syringe pumps generate a continuous pressure gradient (∇P) across the extracellular space. Fluid movement within the porous brain matrix is governed by Darcy's Law:

$$v = -\frac{k}{\mu} \nabla P \quad (5)$$

where v represents interstitial fluid velocity, k is intrinsic tissue hydraulic permeability, and μ is infusate dynamic viscosity.

Under active convection, bulk fluid flow becomes the dominant transport mechanism, substantially reducing the relative importance of molecular size compared with passive diffusion [25, 27]. This changes the relevant design variable from molecular diffusivity to controllable fluid transport and tissue compliance. Small-molecule antivirals, monoclonal antibodies, and viral vectors distribute over centimeter-scale distances within hours rather than weeks. By bypassing microvascular transport barriers and overcoming diffusion limits, CED facilitates targeted fluid distribution across deep cerebral structures.

3 Stereotactic Target Selection, Surgical Trajectories, and Parenchymal Biomechanics

3.1 Spatial Dissemination and Limitations of Unifocal Infusion

The clinical translation of CED has historically focused on neuro-oncology, particularly high-grade gliomas such as glioblastoma multiforme. In localized neoplastic disease, stereotactic catheter placement targets an anatomically circumscribed tumor mass or adjacent resection margins to saturate a focal volume of tissue. Applying this methodology to rabies encephalitis presents a fundamentally different topological problem. Rabies virus propagates trans-synaptically through interconnected neural circuits rather than expanding as a localized

solid mass. Because the pathogen disseminates through continuous brainstem and subcortical pathways, a unifocal infusion site is biophysically and geometrically insufficient: a single cannula cannot establish convective coverage across bilateral thalamic, pontine, and medullary targets.

Effective therapeutic coverage mandates a distributed, multi-catheter stereotactic network. Because baseline parenchymal compliance, inflammatory edema, and local viral loads vary dynamically as encephalitis advances, convective infusion profiles must be evaluated dynamically via co-infusion of magnetic resonance imaging (MRI) surrogate tracers such as gadolinium-DTPA complexes to confirm parenchymal fluid distribution in real time [28].

3.2 Surgical Trajectory Planning and Anatomical Targeting

Target coordinates must be selected based on an integration of viral tropism, surgical accessibility, tissue elastance, and parenchymal microstructure:

- **Supratentorial Targets (Bilateral Thalamic and Striatum):** Pre-coronal burr-hole trajectories orthogonal to the skull surface allow cannulation of the deep thalamic nuclei and basal ganglia while traversing non-eloquent frontal white matter, avoiding the primary motor strip and superior sagittal sinus.
- **Infratentorial Targets (Dorsal Pons and Medulla Oblongata):** Cannulation of the dorsal pons (locus coeruleus, parabrachial complex) and medulla (nucleus ambiguus, dorsal motor nucleus of the vagus) requires suboccipital or trans-cerebellar trajectories planned under intraoperative stereotactic guidance to avoid fourth ventricular entry and transverse sinus injury.

Surgical cannulation of inflamed, encephalitic tissue introduces risks of microvascular shear injury and parenchymal hemorrhage. When volumetric infusion rate (Q) exceeds the tissue's endogenous hydraulic clearance capacity (Q_{drain}), fluid accumulates locally within the extracellular matrix, elevating catheter tip pressure (P_{tip}). Excessive backpressure shears delicate neuropil and induces focal cavitation edema [26, 27]. To prevent mechanical disruption within the medulla and pons, proposed preclinical infusion protocols evaluate stepped micro-ramping, escalating gradually from 0.2 to 0.5 $\mu\text{L}/\text{min}$. Flow rates of 1.0 to 3.0 $\mu\text{L}/\text{min}$ are reserved strictly for larger supratentorial grey matter structures such as the thalamus and putamen.

3.3 Mechanical Divergence: Grey Matter Isotropy versus White Matter Shunting

Grey and white matter diverge mechanically:

- Subcortical grey matter comprises neuronal cell somas and neuropil with an extracellular space width of 15 to 25 nm and a volume fraction (α) of approximately 0.20 [23, 24]. The interstitial space contains an isotropic extracellular matrix rich in hyaluronic acid and chondroitin sulfate proteoglycans, producing uniform hydraulic resistance and generating symmetrical, roughly spherical convective distribution fronts.

Table 1: Target-Region Selection, Neuroanatomical Tropism, and Illustrative Convective Regimes

Anatomical Target	Pathological Tropism & Role	Stereotactic Trajectory	Infusion Flow Regime
Medulla Oblongata (DMNV, Nucleus Ambiguus)	Primary autonomic gating hub; origin of fatal dysautonomia	Infratentorial trans-cerebellar approach	0.2 to 0.5 $\mu\text{L}/\text{min}$; stepped micro-ramp
Dorsal Pons	Locus coeruleus; autonomic and ventilatory relay	Suboccipital retrosigmoid trajectory	0.2 to 0.5 $\mu\text{L}/\text{min}$; real-time MRI monitoring
Bilateral Thalami	Neocortical relay and circuit gating center	Trans-frontal pre-coronal trajectory	1.0 to 3.0 $\mu\text{L}/\text{min}$; spherical convective front
Basal Ganglia / Striatum	Extrapyramidal motor amplification focus	Orthogonal coronal burr-hole approach	1.0 to 2.0 $\mu\text{L}/\text{min}$; boundary avoidance
Cervical Subarachnoid (C3–C5)	Phrenic motor neuron pools; respiratory drive	Percutaneous cervical micro-catheter	Pulsed micro-perfusion (10–20 $\mu\text{L}/\text{min}$)

- White matter behaves anisotropically. Parallel, tightly packed myelinated axon bundles create directional hydraulic channels, with longitudinal hydraulic conductivity along the axonal axis vastly exceeding transverse conductivity. Cannula tips positioned within or adjacent to white matter tracts risk directional hydraulic shunting, channeling infusate along fiber tracts away from targeted subcortical nuclei.
- Reflux along the cannula exterior represents a distinct mechanical failure. When local tip pressure (P_{tip}) exceeds parenchymal yield strength (σ_y), the tissue seal detaches, allowing infusate to escape backward along the insertion tract into the subarachnoid space or ventricles. Mitigating reflux requires stepped-tip cannula geometries combined with controlled micro-infusion ramping protocols [29, 30].

4 The RVNA Architecture: Mechanisms, Fluid Kinetics, and Delivery Modalities

4.1 Core Operational Principles

The RVNA protocol treats symptomatic rabies encephalomyelitis as an active biophysical transport problem requiring coordinated hydrodynamic and molecular operations:

1. **Active Interstitial Convection:** Stereotactically guided, positive-pressure micro-infusion generates continuous bulk fluid flow through the brain parenchyma, substantially reducing the relative importance of molecular size compared with passive diffusion [25, 27].
2. **Selective Regional Cranial Hypothermia:** Closed-loop cranial heat exchange cools the target parenchyma to an illustrative proposed range of 30 to 34 °C in preclinical models to suppress viral polymerase processivity and block

secondary excitotoxic cascades. Systemic active warming maintains core normothermia (36.5 to 37.5 °C), eliminating systemic coagulopathies, myocardial suppression, and refractory cardiac dysrhythmias.

3. **Spatially Stratified Virucidal Neutralization:** Compartmental boundaries dictate antibody geometry. Full-length monoclonal antibodies neutralize free extracellular virions within the bulk interstitial matrix, while single-domain antibodies penetrate the narrow inter-neuronal synaptic cleft to arrest cell-to-cell spread.
4. **Intracellular Transcriptional Arrest:** Synthetic adenosine analogues and sequence-specific oligonucleotides arrest viral RNA-dependent RNA polymerase (RdRp) elongation and recruit endogenous nucleases to degrade viral messenger transcripts without lysing host neurons [31, 37].
5. **Host-Targeted Metabolic Deprivation:** Chemical inhibition of the host mevalonate pathway halts lipid droplet biogenesis, eliminating the structural endomembrane platforms required for lyssavirus envelope assembly and viral budding [32].
6. **Cytoskeletal Stabilization:** Staged micro-infusion of neurotrophic factors repairs damaged neuronal architecture, resolving axonal varicosities, restabilizing disrupted neurofilaments, and restoring physiological transport networks following viral suppression [33].

4.2 Pharmacological Stack and Formulation Chemistry

Co-infusing multiple pharmacological entities into the central nervous system requires rigorous formulation chemistry to prevent precipitation, aggregation, and catastrophic cannula occlusion:

- **Vehicle and Buffer Composition:** All aqueous biologics and hydrophilic small molecules are formulated in sterile, isotonic artificial cerebrospinal fluid (aCSF: 147 mM NaCl, 2.7 mM KCl, 1.2 mM CaCl₂, 0.85 mM MgCl₂, 1.0 mM Na₂HPO₄, pH 7.38–7.42, osmolality 295–305 mOsm/kg).
- **SYN023 (Zamovimab and Mazorelvimab):** Recombinant humanized IgG1 monoclonal antibody cocktail ($M_w \approx 150$ kDa; hydrodynamic radius $R_h \approx 5.5$ nm) comprising non-competing clones zamovimab and mazorelvimab [34, 53]. Evaluated in Phase 2b and Phase 3 randomized, double-blind trials across 1,448 subjects with Category III rabies exposure receiving post-exposure prophylaxis, SYN023 binds two distinct, non-overlapping conformational epitopes on the rabies virus glycoprotein (RABV-G) ectodomain (antigenic sites I and III). Steric binding blocks host receptor attachment (nAChR, NCAM, p75^{NTR}) and halts endocytosis. Due to pronounced steric hindrance, dense extracellular matrix geometry, and trans-synaptic adhesion protein networks within the 20 to 30 nm synaptic cleft, large bivalent IgG1 molecules exhibit severely restricted partitioning into this microenvironment, confining their neutralizing activity to free virions circulating within the bulk interstitial fluid.

- **Biparatopic VHH Nanobodies (Rab-E8/H7):** Engineered camelid single-domain antibody fragments ($M_w \approx 12$ –15 kDa; $R_h \approx 1.5$ –2.0 nm) directed against conserved structural pockets and fusion loops on the RABV-G ectodomain provide picomolar affinity while avoiding steric exclusion [35, 36]. In rodent challenge models, intracerebral micro-infusion of 33 μ g of the biparatopic nanobody Rab-E8/H7 protected mice against lethal rabies challenge [35, 36]. With minimal hydrodynamic radius and low molecular weight, nanobodies partition into the crowded 20 to 30 nm synaptic cleft, intercepting budding progeny at the presynaptic membrane before postsynaptic receptor engagement occurs.
- **Galidesivir (BCX4430 / Immucillin-A):** A synthetic broad-spectrum adenosine C-nucleoside analogue ($M_w = 291.3$ Da) [31, 37]. Host nucleotide kinases phosphorylate the molecule into its active 5'-triphosphate congener, which the viral RNA-dependent RNA polymerase (L-protein complex) incorporates into nascent viral RNA, provoking immediate non-obligate chain termination.
- **Atorvastatin Formulation:** Atorvastatin is lipophilic and virtually insoluble in neutral aqueous saline. To avoid protein aggregation and precipitation when co-administered alongside hydrophilic biologics, atorvastatin is administered via separate enteral/parenteral systemic routes or solubilized with 2-hydroxypropyl- β -cyclodextrin (HP- β -CD) in a dedicated micro-infusion channel. In cell-culture models, atorvastatin inhibits lipid droplet biogenesis via the NDRG1-DGAT1/2 pathway, depriving budding virions of essential structural endomembrane scaffolds [32].
- **2'-MOE ASO Gappers:** Synthetic single-stranded 2'-O-methoxyethyl phosphorothioate chimeric gappers targeted to conserved regions of the lyssavirus nucleoprotein (N) and polymerase (L) mRNAs. The negative-sense genomic RNA core is tightly wrapped by viral N protein (one monomer per nine nucleotides), making intact genomic RNA sterically inaccessible to antisense binding. Gappers bind free viral mRNAs, recruiting endogenous nuclear and cytoplasmic RNase H1 to execute catalytic template cleavage without host cell lysis.

4.3 Protocol Modularity and Dynamic Payload Titration

The therapeutic regimen requires dynamic calibration to match patient-specific physiological constraints, viral genotype, intracranial compliance, and treatment phase:

- In cases involving antigenic divergence (e.g., Phylogroup II lyssaviruses like Lagos bat or Mokola virus that lack SYN023 binding epitopes), full-length IgG1 infusion is halted immediately to avoid futile volume loading. Fluid volume is reallocated immediately to cross-reactive pan-lyssavirus VHH nanobodies, nucleosides, and antisense gappers.
- Volumetric delivery rates are bounded by intracranial elastance. When catheter tip backpressure (P_{tip}) elevates or in-

traoperative MRI reveals peri-cannula fluid pooling, volumetric flow rates (Q) must be reduced immediately. In this volume-restricted state, higher-viscosity biologics are temporarily suspended to prioritize compact small molecules and antisense oligonucleotides.

- The emergence of drug-induced transaminitis, elevated serum creatine kinase, or deteriorating renal function requires prompt cessation of host metabolic modulators, particularly atorvastatin.
- Exogenous neurotrophic factors require strict segregation from the active antiviral phase: infusing BDNF, NT-3, or NGF during active viral transcription accelerates the metabolic activity of infected neurons and forces actin polymerization, feeding trans-synaptic viral spread. Neurotrophic factors are omitted during induction and introduced only following confirmation of viral suppression via serial CSF RT-qPCR.

4.4 Ventricular Delivery Mechanics and Fallback Limitations

Intraparenchymal Convection-Enhanced Delivery (CED) serves as the primary interventional modality. When stereotactic neurosurgical assets are unavailable, an intracerebroventricular (ICV) or intrathecal infusion track functions as a temporary salvage pathway.

Instilling therapeutic agents directly into the cerebrospinal fluid alters physical transport dynamics, introducing three anatomical and physiological barriers:

1. The ventricular wall consists of an ependymal cell monolayer joined by junctional complexes, functioning as a selective physical filter that prevents bulk fluid movement into the brain parenchyma and restricts foreign macromolecules to concentration-dependent diffusive penetration [21].
2. Electrostatic interactions create a surface trap: engineered nanobodies and adjuvant peptides frequently possess cationic surface patches that undergo non-specific electrostatic interactions with polyanionic sialic acid and heparan sulfate proteoglycans on the ependymal surface, capturing payloads along the ventricular lining.
3. Bulk CSF turnover acts as an aggressive hydrodynamic sink. In adult humans, physiological turnover runs between 0.3 and 0.4 mL/min, flushing 430 to 580 mL/day toward arachnoid granulations and dural lymphatics for systemic venous reabsorption [54]. Before foreign macromolecules can diffuse across the ependymal lining into periventricular target nuclei, hydrodynamic clearance sweeps them out of the vault.

4.5 Comparative Transport Modeling: CED versus ICV Delivery

Solute penetration from the ventricular compartment into parenchyma is governed by Fick's Second Law with microvascular elimination:

$$\frac{\partial C}{\partial t} = D^* \nabla^2 C - k_e C \quad (6)$$

where C is local solute concentration, k_e is the elimination rate constant across capillaries and ependyma, and D^* is the effective tissue diffusion coefficient, corrected for extracellular space volume fraction ($\alpha \approx 0.20$) and geometric tissue tortuosity ($\lambda \approx 1.6$) [23]:

$$D^* = \frac{D_0}{\lambda^2} \quad (7)$$

Applying this framework, the RVNA reduced-order transport model uses illustrative parenchymal diffusion coefficients of $D_{IgG}^* \approx 1.0 \times 10^{-7} \text{ cm}^2/\text{s}$ for full-length antibodies and $D_{VHH}^* \approx 6.5 \times 10^{-7} \text{ cm}^2/\text{s}$ for compact single-domain nanobodies.

The Einstein-Smoluchowski equation defines the characteristic diffusion timescale (t) required to traverse a transport distance (L):

$$t \approx \frac{L^2}{2D^*} \quad (8)$$

Under unpressurized diffusion over 1.0 cm, transit requires 57.8 days for IgG and 8.9 days for VHH nanobodies.

In contrast, under positive-pressure CED, fluid transport through the porous brain matrix is governed by Darcy's Law ($v = -(k/\mu)\nabla P$). Convective transit time scales directly with anatomical volume (V_d) and volumetric flow rate (Q):

$$t_{conv} = \frac{V_d}{Q} \quad (9)$$

For a targeted subcortical grey matter structure ($V_d = 1.0 \text{ cm}^3$) infused at an operational flow rate $Q = 3.0 \text{ } \mu\text{L}/\text{min} = 3.0 \times 10^{-3} \text{ cm}^3/\text{min}$:

$$t_{conv} = \frac{1.0 \text{ cm}^3}{3.0 \times 10^{-3} \text{ cm}^3/\text{min}} \approx 333.3 \text{ minutes} \approx 5.5 \text{ hours} \quad (10)$$

Under positive-pressure CED, the molecular-weight dependence of parenchymal distribution is substantially reduced relative to passive diffusion, enabling saturation of a 1.0 cm^3 subcortical target structure within approximately 5.5 hours. This kinetic profile is compatible with the operational window of acute encephalitic rabies.

4.6 Operational Optimization for the ICV Salvage Track

When stereotactic neurosurgical assets are unavailable, clinical execution of the ICV fallback track requires two protocol modifications:

1. Because continuous low-rate gravity infusions wash out rapidly through arachnoid granulations, precision syringe pumps must be programmed to deliver automated high-frequency pulsed micro-infusions at an illustrative rate of 10 to 20 $\mu\text{L}/\text{min}$ for 30 seconds every 10 minutes. These micro-bursts create localized hydrostatic pressure eddies within ventricular compartments. The induced fluid turbulence disrupts the unstirred ependymal boundary layer, promoting solute partitioning into the ventricular wall before bulk CSF turnover clears the instilled volume.
2. Concurrently, dosing volumes must be compartmentalized and re-allocated. Unpressurized IgG cannot penetrate deep

Table 2: Transport Regime Comparison: Unpressurized Ventricular Diffusion (ICV) versus Convective Bulk Advection (CED)

Transport Parameter	Passive Ventricular Diffusion (ICV Track)	Convection-Enhanced Delivery (CED Track)
Driving Mechanism	Concentration Gradient (∇C)	Hydrostatic Pressure Gradient (∇P)
Mass Dependency	Inversely scaled to M_w and R_h	Bulk advection dominates; size dependency substantially reduced
Transport Mode	Unpressurized Brownian dispersion	Forced bulk interstitial advection
Transit Time: 1.0 cm (IgG)	57.8 days (Model-Derived Theoretical)	≈ 5.5 hours (Illustrative Model-Derived)
Transit Time: 1.0 cm (VHH)	8.9 days (Model-Derived Theoretical)	≈ 5.5 hours (Illustrative Model-Derived)
Translational Feasibility	Diffusion-limited; incompatible with acute rabies	Therapeutically viable within the acute clinical window

parenchyma; continuing standard SYN023 infusions simply pools protein periventricularly, adding fluid volume without target engagement. The protocol therefore cuts SYN023 dosing by 50%, transferring the spare volume directly to compact, rapidly diffusing payloads: biparatopic VHH nanobodies ($D^* \approx 6.5 \times 10^{-7} \text{ cm}^2/\text{s}$) and galidesivir ($M_w = 291.3 \text{ Da}$).

5 The Translational Collapse of Systemic Transcytosis Shuttles in Human-Scale Infection

5.1 Hydrodynamic Mass-Balance and Visceral Mass Sinks

Delivering macromolecules across the blood-brain barrier (BBB) via receptor-mediated transcytosis (RMT) or adsorptive-mediated transcytosis (AMT) faces prohibitive physiological hurdles [38, 39, 40]. In an elegant rodent proof-of-concept study, Ren et al. demonstrated that a cationic SynB1-conjugated chimeric antibody cocktail crossed the mouse BBB and achieved an 80% survival rate when treatment began at 5 days post-infection [38]. The mouse result establishes that BBB-shuttle engineering can be therapeutically consequential; it does not establish that the same transport architecture provides sufficient, spatially uniform exposure in established human symptomatic disease. Translating peptide-shuttle conjugates to adult humans is constrained by peripheral organ distribution. As Ren et al. observed, SynB1 accumulates prominently in liver and kidney tissue due to its charge-mediated cell-penetration mechanism [38].

In humans, visceral vascular surface area exceeds the cerebral microvascular footprint by roughly two orders of magnitude. Systemic administration exposes shuttles targeting transferrin receptor 1 (TfR1) or cationic motifs to large peripheral receptor sinks before the constructs reach cerebral capillaries [39, 40]. As Pardridge established, unassisted paracellular transit across the intact human BBB excludes $> 99.9\%$ of circulating large molecules [40]. In an adult vascular volume (5 L), clearance by visceral capillary beds overwhelmingly outcompetes cerebral extraction:

$$\frac{dC_p}{dt} = - \left(\frac{CL_{\text{visceral}} + CL_{\text{BBB}}}{V_d} \right) C_p(t) \quad (11)$$

This vast peripheral extraction sink precludes meaningful therapeutic brain exposure. Endogenous ligands outcompete synthetic shuttles for target receptors, while attempting to saturate peripheral clearance via dose escalation risks severe target-

mediated drug disposition toxicity, including reticulocyte lysis and hepatic injury.

5.2 Ultrastructural Divergence and Endothelial Transit Kinetics

Luminal capture does not ensure parenchymal delivery. Even if a fractional payload engages luminal brain capillaries, intracellular trafficking across human brain microvascular endothelial cells (BMECs) faces a degradation-dominated sorting problem. As Yu et al. demonstrated with TfR1 antibodies, high receptor affinity routes endocytosed complexes predominantly toward lysosomal degradation rather than abluminal transcytotic egress [39]. The affinity optimum represents a design conflict: high affinity promotes luminal capture but reduces basolateral sorting efficiency, while low affinity favors abluminal release but impairs initial capture in high-shear human capillaries.

Endothelial vesicular transit operates as a coupled advection-diffusion-reaction phenomenon. In our reduced-order transport model, the probability of intact vesicular transcytosis (η_{trans}) across the endothelial spatial domain $x \in [0, \delta_{\text{EC}}]$ scales exponentially against transit delay and lysosomal interception:

$$\eta_{\text{trans}} \approx \exp \left(-\delta_{\text{EC}} \sqrt{\frac{k_{\text{lys}}}{D_{\text{ves}}} + \left(\frac{v_{\text{ves}}}{2D_{\text{ves}}} \right)^2} + \frac{v_{\text{ves}} \delta_{\text{EC}}}{2D_{\text{ves}}} \right) \quad (12)$$

where δ_{EC} denotes endothelial cross-sectional thickness, D_{ves} is the vesicular diffusion coefficient, v_{ves} represents forward transcytotic velocity, and k_{lys} represents the rate constant for lysosomal fusion.

Human endothelial architecture amplifies this trap. Human BMECs present a thicker cross-sectional profile ($\delta_{\text{EC}, \text{human}} > \delta_{\text{EC}, \text{murine}}$) and much denser pericyte coverage. This extended path length prolongs vesicular transit. Longer transit gives lysosomal enzymes more time to intercept endocytosed payloads. Whenever forward vesicular velocity stalls ($v_{\text{ves}} \rightarrow 0$), lysosomal degradation (k_{lys}) dictates payload fate, leaving little material intact for abluminal release.

5.3 Post-Egress Interstitial Transport Failure

Even when an antibody conjugate traverses the brain capillary endothelium, it encounters a second transport barrier within the parenchyma. Extravasated molecules exit the basolateral membrane into the perivascular Virchow-Robin space, where subsequent transport across the neuropil relies almost entirely on pas-

sive interstitial diffusion:

$$\frac{\partial C_{\text{tissue}}}{\partial t} = \frac{D_0}{\lambda^2} \nabla^2 C_{\text{tissue}} - k_{\text{elim}} C_{\text{tissue}} \quad (13)$$

where λ quantifies extracellular matrix tortuosity ($\lambda \approx 1.6$) and k_{elim} reflects local microvascular clearance based on the porous media transport framework [23, 24]. In the RVNA transport model, an illustrative effective diffusion coefficient of $D_{\text{eff}}^* \approx 1.0 \times 10^{-7} \text{ cm}^2/\text{s}$ parameterizes the severe steric hindrance experienced by a canonical 150 kDa immunoglobulin conjugate.

Passive diffusion is inadequate for macroscopic transport. Continuous capillary resorption attenuates the advancing concentration gradient, confining extravasated antibodies to a narrow perivascular zone typically less than 100 micrometers from the capillary surface. Symptomatic rabies requires therapeutic coverage across centimeter-scale subcortical and brainstem nuclei. Passive gradients cannot achieve this within the acute clinical window. Convection-enhanced delivery (CED) resolves the spatial barrier by replacing passive concentration gradients with active, continuous hydrostatic bulk convection.

5.4 Kinetic Mismatch with Disease Progression

Rabies virions exploit active, dynein-driven retrograde axonal transport machinery to rapidly colonize brainstem, thalamic, and medullary autonomic nuclei [9, 41]. Let t_{sx} denote clinical rabies onset, marking established neuroinvasion of the brainstem and deep cranial nuclei. In our analytical transport framework, survival probability $S(t)$ governed by systemic transcytosis shuttles is modeled by an illustrative piecewise boundary:

$$S(t) = \begin{cases} 1.00 & \text{if } t \leq t_{\text{inoculation}} \quad (\text{pre-invasion}) \\ 0.10 & \text{if } t_{\text{entry}} < t < t_{\text{sx}} \quad (\text{spinal/medullary transit}) \\ 0.00 & \text{if } t \geq t_{\text{sx}} \quad (\text{established clinical disease}) \end{cases} \quad (14)$$

The kinetic mismatch is decisive within the modeled clinical window. Passive extravasation and vesicular transcytosis require days or weeks to establish minimal therapeutic tissue thresholds. Fatal autonomic collapse routinely occurs within 3 to 10 days of clinical onset. Transport models indicate that systemic shuttles cannot achieve therapeutic parenchymal concentrations within the operational window defined by $t \geq t_{\text{sx}}$. Direct intraparenchymal CED circumvents the systemic mass sink, eliminating vascular transit delays and establishing convective distribution directly within viral replication centers.

6 Developmental Neurobiology and the Post-Clearance Recovery Bottleneck

6.1 Pathological Substrate: Cytoskeletal Decimation and Synaptic Pruning

Virological clearance in the central nervous system does not ensure clinical recovery. Rabies neuropathology is model-dependent: suckling-mouse systems demonstrate substantial apoptosis, whereas mature experimental systems more often exhibit severe neuronal-process degeneration with comparatively limited somatic loss [52, 33, 42]. In adult animal models and

human post-mortem studies, fatal dysautonomia results primarily from neuronal process degeneration and synaptic dysfunction rather than widespread somatic destruction.

Viral proliferation causes marked depolymerization of filamentous actin (F-actin) and microtubule networks [6, 33]. The viral phosphoprotein (P protein) subverts host cytoskeletal binding proteins and hijacks dynein motor complexes to drive retrograde axonal transport. Concurrently, RABV infection induces oxidative stress and lipid peroxidation within axons, precipitating the formation of pathological axonal swellings and widespread dendritic beading. In mature experimental systems, histological assessments demonstrate that process degeneration predominates over somatic lysis, confirming that clinical dysautonomia stems from structural and functional disconnection of viable somas [33].

Following viral clearance, the surviving neural architecture remains heavily pruned. The tissue consists of viable somas stripped of dendritic spines and disconnected by fragmented axonal tracts, producing severe functional dis-integration of subcortical and brainstem networks.

6.2 Developmental Allometry: Neuroplasticity and the Pediatric Epigenetic Divide

Reconstructing this injured synaptic architecture reveals an allometric divide. Brain-derived neurotrophic factor (BDNF) serves as the primary regulator of dendritic spine density, neurotransmission, and experience-dependent neuroplasticity through tyrosine kinase receptor B (TrkB) activation [43, 44]. Pediatric populations possess elevated baseline synaptic density and active neurogenic reservoirs, providing greater structural potential to re-establish damaged brainstem networks.

Neuroinflammation constrains this pediatric advantage. In this proposed neurorepair framework, pediatric plastic potential is heavily constrained by the transcriptional response of the immature neurovascular unit to acute viral neuroinflammation. We hypothesize that severe encephalitic inflammation suppresses endogenous neuroplastic repair by disrupting physiological BDNF expression. MicroRNAs including miR-1 and miR-10b directly repress BDNF expression through specific regulatory target sites in the 3'-UTR of the BDNF transcript [45]. In our neurorepair model, rabies-induced neuroinflammatory signaling and associated cytokine cascades drive pathological up-regulation of these repressive microRNAs alongside promoter DNA methylation, transcriptionally silencing endogenous repair pathways [6, 46]. Adult tissue, possessing consolidated epigenetic methylation profiles, resists sudden inflammatory epigenetic shifts, yet its maximal re-innervation capacity remains inherently limited.

6.3 The Virological Control Gate Constraint: Kinetic Risks of Premature Synaptogenesis

Bypassing endogenous epigenetic silencing via direct parenchymal micro-infusion of exogenous neurotrophins (such as recombinant BDNF, Nerve Growth Factor, or Neurotrophin-3) introduces a critical kinetic hazard. RABV infection relies on a dynamic actin and microtubule network for clathrin-mediated cellular entry, transcription, and progeny egress [6, 47]. Activating TrkB or TrkC pathways via exogenous growth factors stimulates

rapid actin polymerization and neurite outgrowth. The proposed gate is therefore designed around a mechanistic risk: stimulating structural plasticity while residual virus remains could increase the availability of cellular transport machinery required for viral spread. In preclinical systems, chemical inhibitors of actin polymerization demonstrate dose-dependent suppression of RABV proliferation [47]. Forcing premature synaptogenesis before complete virological clearance risks accelerating trans-synaptic dissemination.

Interventional protocols must implement a hard virological clearance gate. In the proposed framework, this gate is defined mechanically as a piecewise operational boundary governing viral RNA concentration in the cerebrospinal fluid (C_{viral}) against the quantitative real-time PCR limit of detection (“LOD”):

$$\Gamma(t) = \begin{cases} 0 & \text{if } C_{viral}(t) \geq \text{LOD} \\ 1 & \text{if } \frac{1}{\Delta t} \int_{t-\Delta t}^t C_{viral}(\tau) d\tau < \text{LOD} \end{cases} \quad (15)$$

where $\Gamma(t) = 0$ halts exogenous neurotrophin infusion and $\Gamma(t) = 1$ initiates synaptic reconstruction, with Δt representing a continuous, multi-assay temporal verification window. Any structural repair initiated while $\Gamma(t) = 0$ actively feeds viral trans-synaptic transport by generating active microtubule and F-actin platforms. Neuroplastic reconstruction must remain strictly downstream of verified viral sterilization.

Box 1 | Key Transport and Clearance Formulations

1. Fickian Diffusion Timescale (Unpressurized CSF):

$$t \approx \frac{L^2}{2D^*} = \frac{L^2 \lambda^2}{2D_0} \implies t_{1.0 \text{ cm, IgG}} \approx 57.8 \text{ days}$$

2. Darcy Advection Timescale (Convection-Enhanced Delivery):

$$v = -\frac{k}{\mu} \nabla P, \quad t_{conv} = \frac{V_d}{Q} \implies t_{1.0 \text{ cm}^3 \text{ nucleus}} \approx 5.5 \text{ hours}$$

3. Virological Control Gate for Neurotrophin Activation:

$$\Gamma(t) = \begin{cases} 0 & \text{if } C_{viral}(t) \geq \text{LOD} \\ 1 & \text{if } \frac{1}{\Delta t} \int_{t-\Delta t}^t C_{viral}(\tau) d\tau < \text{LOD} \end{cases}$$

4. Synaptic Collapse Threshold for Autonomic Failure:

$$\Phi_{synaptic}(t) = \frac{N_{intact}(t)}{N_{baseline}}, \quad \text{Fatal threshold: } \Phi_{synaptic} < \Phi_{crit} \approx 0.15$$

7 RVNA Drug Compendium, Phased Protocols, and Temporal Dynamics

7.1 Phased Interventional Schedule

To ensure biological safety and prevent pharmaceutical incompatibilities, drug delivery is organized into three chronologically gated phases:

- **Phase I (Immediate Advective Arrest):** Stereotactic CED of VHH nanobodies (Rab-E8/H7) and galidesivir into the dorsal pons and medulla oblongata, accompanied by concurrent cervical subarachnoid micro-perfusion, to immediately arrest trans-synaptic circuit transmission and terminate viral transcription.

- **Phase II (Bulk Clearance and Morphogenesis Deprivation):** Supratentorial CED of SYN023 and 2'-MOE ASO gapmers into the thalami and striatum, accompanied by systemic/enteral atorvastatin administration to deplete host lipid droplet budding scaffolds.
- **Phase III (Gated Neuroplastic Reconstruction):** Targeted micro-infusion of recombinant BDNF, NT-3, and NGF initiated strictly after satisfying the virological clearance gate ($\Gamma(t) = 1$) to reverse axonal beading and rebuild synaptic connections.

7.2 Pharmacological Profiling and Compartmental Assignment

- **SYN023 IgG1 Cocktail (150 kDa):** Formulated in isotonic aCSF (pH 7.4). Clears extracellular virions in the bulk interstitial space. Sterically excluded from the synaptic cleft.
- **Biparatopic VHH Nanobodies (Rab-E8/H7, 15 kDa):** Formulated in isotonic aCSF. High-affinity neutralizer of conserved glycoprotein fusion loops. Partitions into the 20 to 30 nm synaptic cleft to block cell-to-cell spread.
- **Galidesivir (BCX4430, 291 Da):** Adenosine C-nucleoside analogue dissolved in aCSF. Readily taken up by neuronal somata and phosphorylated to active triphosphate, inducing non-obligate viral RdRp chain termination.
- **Atorvastatin:** Administered enterally or via dedicated HP- β -CD vehicle. Depletes mevalonate intermediates, suppressing lipid droplet biogenesis and halting virion assembly platforms.
- **2'-MOE ASO Gapmers:** Phosphorothioate chimeric gapmers targeting conserved regions of N and L transcripts. Recruits endogenous RNase H1 to cleave viral mRNAs without cytolysis.
- **Neurotrophin Stack (BDNF, NT-3, NGF):** Recombinant growth factors infused post-clearance to restabilize neurofilaments and drive dendritic arborization.

7.3 Compartmentalized Neutralization Kinetics: SYN023 and VHH Constructs

Spatial compartmentalization is dictated by physical transport boundaries. Full-length IgG1 antibodies cannot operate uniformly throughout all neural microenvironments. SYN023 combines two non-competing humanized monoclonal clones (zamorvimab and mazorelvimab) raised against non-overlapping epitopes on the glycoprotein ectodomain. In Phase 2b and Phase 3 trials involving 1,448 Category III post-exposure patients, peripheral safety and virucidal potency were verified [34]. Inside the brain, however, biophysical transport constraints dominate. At 150 kDa, SYN023 is assigned an illustrative effective diffusion coefficient (D_{eff}^*) of roughly $1.0 \times 10^{-7} \text{ cm}^2/\text{s}$ in the RVNA transport calculation. The inter-neuronal junction worsens these limits: crowded with proteoglycan meshworks, adhesion filaments, and tortuous geometries, the 20 to 30 nm synaptic cleft sterically excludes bivalent immunoglobulins. SYN023

Table 3: RVNA Investigational Drug Compendium and Delivery Specifications

Agent	Molecular Class		Mechanism of Action	Delivery Route	Formulation Vehicle
SYN023	Recombinant humanized IgG1		Neutralizes RABV-G sites I and III	Supratentorial CED	Isotonic aCSF (pH 7.4)
Rab-E8/H7	Biparatopic nanobody	VHH	Picomolar cleft neutralizer	Infratentorial / Spinal CED	Isotonic aCSF (pH 7.4)
Galidesivir	Adenosine C-nucleoside		Viral RdRp chain termination	Brainstem / Thalamic CED	Isotonic aCSF (pH 7.4)
Atorvastatin	Lipophilic statin		Inhibits lipid droplet assembly	Enteral / Systemic IV	Separate lipid / HP- β -CD vehicle
2'-MOE Gapmers	Phosphorothioate ASO		RNase H1 catalytic mRNA cleavage	Supratentorial CED	Isotonic aCSF (pH 7.4)
BDNF / NT-3 / NGF	Recombinant retrophins	neu-	TrkB/TrkC axonal stabilization	Delayed CED ($\Gamma(t) = 1$)	Isotonic aCSF with human albumin

is therefore restricted to clearing free virions drifting within the open interstitial bath.

Arresting cell-to-cell spread across synaptic junctions requires neutralizing agents with a vastly reduced steric footprint. Camelid single-domain antibody fragments ($M_w \approx 12\text{--}15$ kDa) provide picomolar affinity while maintaining minimal steric volume. In preclinical rodent models, intracerebral administration of 33 μg of the biparatopic construct Rab-E8/H7 protected mice against lethal challenge [35, 36], demonstrating that single-domain antibodies can intercept active infection within nervous tissue. However, this neutralization capacity remains unproven in human clinical trials. Driven by positive-pressure Convection-Enhanced Delivery (CED), the compact hydrodynamic profile and reduced steric footprint of these 15 kDa nanobodies enable favorable partitioning into the crowded interneuronal junction to intercept budding virions prior to postsynaptic receptor engagement and arrest retrograde trans-synaptic transit along connected neural circuits.

7.4 Intracellular Arrest: Galidesivir and Antisense Oligonucleotides

Interrupting intraneuronal viral transcription relies on targeted polymerase chain termination. Galidesivir (BCX4430 / Immucillin-A) is an adenosine C-nucleoside analogue whose active 5'-triphosphate metabolite acts as an alternative substrate for the viral RNA-dependent RNA polymerase (L-protein complex), causing premature non-obligate chain termination. Preclinical studies demonstrated potent broad-spectrum activity against filoviruses and other non-segmented negative-sense RNA viruses [31]. Human clinical studies have evaluated its safety and pharmacokinetics [37], but therapeutic efficacy against clinical human lyssavirus encephalitis remains unverified.

To enforce transcriptional arrest, the protocol integrates synthetic single-stranded 2'-O-methoxyethyl (2'-MOE) phosphorothioate chimeric gapmer oligonucleotides targeted to conserved sequences within the lyssavirus nucleoprotein (N) and polymerase (L) transcripts. Intracellular hybridization creates an RNA-DNA duplex within the central deoxynucleotide gap, recruiting endogenous RNase H1 to execute catalytic template cleavage and suppress viral mRNA translation.

7.5 Metabolic Extinction of Viral Assembly: Atorvastatin

Lyssavirus morphogenesis requires host structural endomembranes for virion assembly and budding. Empirical analyses in cell-culture and preclinical rodent models show that rabies virus recruits host intracellular lipid droplets as platforms for matrix protein alignment and glycoprotein envelopment via the NDRG1-DGAT1/2 pathway [32].

Atorvastatin inhibits 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase, chemically depleting mevalonate pathway intermediates. By suppressing lipid droplet biogenesis, atorvastatin treatment in vitro deprives budding virions of essential structural assembly scaffolds without provoking host cell cytolysis [32].

7.6 Neurotrophic Repair and Cytoskeletal Stabilization

Absolute viral clearance yields a pruned neuronal matrix characterized by F-actin depolymerization and profound synaptic regression [33]. Reconstructing this ultrastructural damage via exogenous neurotrophic factors (BDNF, NT-3, NGF) carries severe kinetic hazards if initiated prematurely. Because rabies virus subverts host actin polymerization for endocytosis and trans-synaptic transit, activating TrkB or TrkC pathways prior to absolute virological sterilization supplies residual virions with active cytoskeletal transport networks [6, 47].

Structural repair is governed by a strict molecular gate. Exogenous neurotrophin administration (I_{neuro}) is mathematically constrained by the viral RNA concentration in the cerebrospinal fluid (C_{viral}) plotted against the quantitative real-time PCR limit of detection ("LOD"):

$$I_{neuro}(t) = \begin{cases} 0 & \text{if } C_{viral}(t) \geq \text{LOD} \\ 1 & \text{if } \frac{1}{\Delta t} \int_{t-\Delta t}^t C_{viral}(\tau) d\tau < \text{LOD} \end{cases} \quad (16)$$

Following continuous multi-assay verification of viral clearance ($I_{neuro} = 1$), targeted parenchymal micro-infusion of recombinant Neurotrophin-3 and Nerve Growth Factor is executed to promote microtubule stabilization, reverse axonal beading, and drive dendritic arborization [33, 43, 44].

8 Compassionate Usage Directive and FDA Expanded Access Framework

8.1 Regulatory Classification and Framework

The pharmacological armamentarium described in the RVNA framework is unapproved and strictly investigational for this indication. Deploying unapproved biologicals and repurposed small molecules in an individual patient setting falls under the expanded access provisions of Title 21 of the Code of Federal Regulations (21 CFR Part 312, Subpart I, §§ 312.300–312.320) [48]. Symptomatic rabies may meet the expanded-access criteria when the applicable regulatory conditions are satisfied, specifically as an immediately life-threatening condition devoid of satisfactory approved therapeutic alternatives (21 CFR 312.305). This regulatory pathway exists to enable emergency clinical rescue under institutional and federal oversight rather than commercial data collection.

Box 2 | Emergency Regulatory Trajectory (21 CFR Part 312)

Step 1: Verbal eIND Authorization (21 CFR 312.310(d))

Contact FDA Emergency Operations for immediate verbal clearance, waiving the standard 30-day review period.

Step 2: Emergency Informed Consent Exception (21 CFR 50.23)

Incapacitated patient and unavailable LAR certified by independent physician in clinical record.

Step 3: Immediate Stereotactic Surgical Intervention

Catheter placement, micro-infusion ramping, and drug deployment proceed without delay.

Step 4: Post-Intervention Regulatory Filings

- Notify IRB within 5 working days (21 CFR 56.104(c)).
- Submit Form FDA 3926 and clinical summaries within 15 working days (21 CFR 312.310(d)(2)).

8.2 Emergency IND (eIND) Activation Kinetics

Clinical rabies outpaces standard written filings. Under 21 CFR 312.310(d), clinicians can obtain emergency IND (eIND) authorization verbally via telephone or secure electronic messaging with FDA officials, permitting immediate drug dispatch and infusion. Physicians secure a Letter of Authorization (LOA) referencing manufacturer IND safety files or submit supporting chemistry, manufacturing, and control data directly. The standard 30-day waiting period (21 CFR 312.40(b)) governs formal commercial trials, not emergency authorizations under § 312.310(d).

8.3 Exemption Mechanics and Post-Infusion Compliance

Emergency administration of the RVNA stack qualifies for exemption from prior IRB approval under 21 CFR 56.104(c). The exemption is not a waiver of accountability: it shifts the compliance obligation to the post-intervention window. The investigator must notify the IRB within 5 business days of initiating the infusion. A completed Form FDA 3926, together with manufacturer LOAs, must reach the FDA within 15 working days of the initial verbal authorization, pursuant to 21 CFR 312.310(d)(2) [49]. A written clinical outcome summary, including all adverse events, is also required.

8.4 Informed Consent Constraints

Informed consent from the patient or a legally authorized representative (LAR) is mandated under 21 CFR Part 50. Symptomatic rabies encephalomyelitis destroys cognitive capacity

rapidly. When the patient cannot consent, time precludes identifying an LAR, and no alternative intervention exists, the investigator invokes the emergency exception under 21 CFR 50.23(a) and (c). An independent physician not involved in the protocol must prospectively certify these conditions in writing within the medical record, with documentation submitted to the IRB within 5 working days.

9 Biomechanical Delivery Limitations and Neuropathological Failure Modes

9.1 Biomechanical Delivery Failures: Reflux, Shunting, and Compliance Limits

Convective delivery hinges on maintaining an intact mechanical seal at the cannula tip [26, 27]. If local tip pressure (P_{tip}) exceeds parenchymal yield strength (σ_y), the tissue seal detaches. Infusate escapes backward along the insertion tract, venting uselessly into the subarachnoid space or ventricles. In experimental studies, uncontrolled reflux reduced parenchymal distribution volume (V_d) by up to 85% [25, 26]. When reflux occurs, convective saturation of target nuclei collapses.

Hydraulic shunting is a distinct but equally significant failure mode. Cannula tips placed within or adjacent to myelinated white matter tracts (the internal capsule, pontine projection tracts) encounter directional corridors of low hydraulic resistance. Infusate flows preferentially along the fiber plane rather than saturating the isotropic grey matter somas of the intended target nucleus, reducing therapeutic coverage of the target zone.

Volume limits are set by intracranial elastance. When infusion rate (Q) exceeds endogenous fluid resorption and vascular clearance (Q_{drain}), extracellular fluid accumulates:

$$\Delta V = \int_0^t (Q - Q_{drain}) d\tau \quad (17)$$

Rising interstitial pressure produces vasogenic edema, microvascular compression, and parenchymal ischemia. In encephalitic tissue, cytotoxic swelling contracts the extracellular volume fraction (α) from a physiological 0.20 to values below 0.08 [23]. This elevates hydraulic resistance, drops the mechanical disruption threshold, and precipitates ICP spikes that force premature cessation of the infusion.

Box 3 | Parenchymal Infusion Failure Boundaries

1. Cannula Reflux Boundary:

$P_{tip} > \sigma_y \implies$ tissue detachment; fluid tracks back along cannula tract into subarachnoid space or ventricles (up to 85% loss in V_d).

2. Directional Shunting Corridor:

Cannula placed near myelinated white matter tracts; infusate channels preferentially along axon planes away from target nuclei.

3. Intracranial Elastance Limit:

$Q > Q_{drain} \implies$ interstitial fluid accumulation; cytotoxic edema drops $\alpha < 0.08$, precipitating acute ICP spikes.

9.2 Anatomical Sequestration: Spinal and Autonomic Sinks

Stereotactic CED remains anatomically localized to diencephalic and brainstem structures. Rabies virus enters the central axis through motor endplates and sensory nociceptors, replicating within dorsal root ganglia and spinal grey matter before

ascending to subcortical amplification hubs [9, 41]. Cannulating the pons and thalamus leaves these spinal foci untouched.

Hydrodynamic forces compound this anatomical isolation: bulk CSF turnover sweeps solutes rostrally toward arachnoid granulations and dural lymphatics rather than driving penetration caudally along the spinal axis [54]. Spinal motor neurons and peripheral autonomic plexuses (including cardiac and celiac ganglia) receive negligible drug exposure, continuing to shed virions centrifugally into visceral organs. Unchecked spinal infection can retrograde re-seed cleared brainstem circuits or destroy phrenic motor pools, provoking terminal ventilatory arrest independent of intracranial clearance.

9.3 Neuropathological Irreversibility: The Point of No Return

Virological clearance and clinical survival are not the same outcome. In adult animal models and human clinical disease, rabies lethality arises primarily through autonomic circuit failure rather than immediate pan-somatic destruction [15, 33]. The virus strips dendritic spines, depolymerizes F-actin, and destabilizes tubulin architecture across brainstem autonomic regulatory nuclei. Neuronal somas survive, while their synaptic connectivity and axonal projections undergo severe structural degeneration.

In the reduced-order biophysical model, $\Phi_{crit} \approx 0.15$ defines the proposed structural failure boundary, parameterizing synaptic process integrity across the dorsal motor nucleus of the vagus, nucleus ambiguus, and rostral ventrolateral medulla:

$$\Phi_{synaptic}(t) = \frac{N_{intact\ processes}(t)}{N_{baseline}} \quad (18)$$

Below this critical threshold ($\Phi_{synaptic} < \Phi_{crit} \approx 0.15$), autonomous cardiac and respiratory coordination collapses irreversibly. Survival probability under the proposed operational model:

$$P(\text{Survival} | \text{Clearance}) = \begin{cases} > 0.70 & \text{if } \Phi_{synaptic} \geq \Phi_{crit} \approx 0.15 \\ 0.00 & \text{if } \Phi_{synaptic} < \Phi_{crit} \end{cases} \quad (19)$$

Below that threshold, clearance of viral RNA from the cerebrospinal fluid does not alter the clinical trajectory. Denuded brainstem somas cannot sustain vasomotor tone, cardiac rhythm, or respiratory drive. The clinical course progresses to refractory circulatory collapse, neurogenic myocardial injury, malignant dysrhythmias, and asystole. In this structural failure state, protocol deployment after extensive circuit disruption cannot restore functional autonomic output.

9.4 Antigenic Escape, Neuroinflammatory Lysis, and Mechanical Failures

- **Antigenic Escape:** Monoclonal targeting introduces profound strain-specific vulnerability. SYN023 targets two conformational epitopes on the RABV-G ectodomain, but this structural precision represents an inherent point of failure: single point mutations at key contact residues (positions 333, 336, or 338 in site III) drastically erode neutralization affinity [50, 51]. Phylogroup II and III lyssaviruses (e.g., Mokola and Lagos bat viruses) diverge even further, lacking these epitopes entirely. Against phylogenetically divergent lyssaviruses, monoclonal neutralization is

severely compromised or abolished, shifting the burden of virucidal coverage onto broad-spectrum nucleosides and antisense oligonucleotides.

- **Neuroinflammatory Lysis:** Rapid intracellular viral lysis carries severe secondary neuroinflammatory hazards. Enzymatic cleavage of viral RNA by oligonucleotides and nucleoside analogues releases substantial genomic and ribonucleoprotein debris directly into the parenchymal neuropil. This debris triggers microglia and astrocytes via TLR3, TLR7, and TLR8 receptors. Inside an unyielding skull, neuroinflammatory cascades quickly provoke secondary edema, microvascular collapse, and accelerated parenchymal injury.
- **Hardware Complications:** Small-gauge micro-cannulas (internal diameter $\leq 100\ \mu\text{m}$) remain vulnerable to protein aggregation, cellular debris accumulation, and tip thrombosis. Catheter insertion through hyperemic, inflamed parenchyma carries a 2% to 5% incidence of tract hemorrhage in published stereotactic series [27]. Small-bore neurocatheters require strict infusion-rate control to avoid reflux and mechanical seal detachment [29, 30]. Long-term externalized cannula systems introduce the additional risk of ascending bacterial ventriculitis and secondary meningitis.

10 Global Work Acknowledgment and Declarations

The RVNA architecture synthesizes findings across neurovirology, microvascular physiology, and neurosurgical bioengineering into a unified interventional framework. The neurovirologists who mapped lyssavirus retrograde transport and innate immune evasion established why passive systemic interventions fail, defining the biophysical requirements for parenchymal clearance. The engineers and pharmacologists who characterized positive-pressure CED fluid dynamics, single-domain antibodies, and antisense gapmers developed the tools that make targeted CNS delivery possible. We have assembled these empirical components into a structured protocol. The theoretical proposal presented here is directly derivative of that published literature. Translating these technologies into clinical interventions for encephalitic infections relies directly on the experimental data, therapeutic failures, and biological insights established across the international research community.

Funding. No external funding was received for this work.

Conflict of Interest Statement. The authors declare no competing financial or commercial interests.

Regulatory and Clinical Research Disclaimer. The RVNA Protocol constitutes an investigational systems architecture and theoretical biophysical engineering proposal for post-exposure management of lyssavirus encephalitis. It does not represent an established clinical treatment, validated medical standard of

care, or prescriptive medical advice. The multi-drug combinations, stereotactic delivery modalities, micro-cannula flow parameters, pharmacokinetic calculations, dosing regimens, and computational gating models delineated herein are derived from preclinical literature, mathematical modeling, and engineering assumptions. They have not been demonstrated safe or efficacious in human clinical trials for encephalitic rabies. Any clinical translation, compassionate use, or expanded access deployment (e.g., pursuant to 21 CFR Part 312, 21 CFR 312.310, and 21 CFR 56.104) requires independent institutional ethics approval, formal regulatory authorization from competent national authorities, validated pharmaceutical supply chains, and specialized neurosurgical facilities.

References

- [1] Fooks, A. R. et al. Rabies. *Nat. Rev. Dis. Primers* **3**, 17091 (2017). DOI: [10.1038/nrdp.2017.91](https://doi.org/10.1038/nrdp.2017.91).
- [2] World Health Organization. *WHO Expert Consultation on Rabies: Third Report*. (WHO Technical Report Series 1012, Geneva, 2018). WHO Publications.
- [3] Hampson, K. et al. Estimating the global burden of endemic canine rabies. *PLoS Negl. Trop. Dis.* **9**, e0003709 (2015). DOI: [10.1371/journal.pntd.0003709](https://doi.org/10.1371/journal.pntd.0003709).
- [4] Kim, Y. & Donnelly, C. A. Substantial under-detection of human rabies cases in the United States. *Commun. Health* **1**, 23 (2026). DOI: [10.1038/s44528-026-00031-4](https://doi.org/10.1038/s44528-026-00031-4).
- [5] World Health Organization. Rabies vaccines: WHO position paper, April 2018. *Wkly. Epidemiol. Rec.* **93**, 201–220 (2018). WHO Position Paper.
- [6] Sun, C. et al. Biological barriers underlying the extremely high fatality of symptomatic rabies: neuroinvasion, immune evasion, and blood-brain barrier restriction. *Front. Immunol.* **17**, 1904824 (2026). DOI: [10.3389/fimmu.2026.1904824](https://doi.org/10.3389/fimmu.2026.1904824).
- [7] Rao, A. K. et al. Use of a modified preexposure prophylaxis vaccination schedule to prevent human rabies: recommendations of the Advisory Committee on Immunization Practices, United States, 2022. *MMWR Morb. Mortal. Wkly. Rep.* **71**, 619–627 (2022). DOI: [10.15585/mmwr.mm7118a2](https://doi.org/10.15585/mmwr.mm7118a2).
- [8] Fishbein, D. B. et al. Administration of human diploid-cell rabies vaccine in the gluteal area. *N. Engl. J. Med.* **318**, 124–125 (1988). DOI: [10.1056/NEJM198801143180219](https://doi.org/10.1056/NEJM198801143180219).
- [9] Gluska, S. et al. Rabies virus hijacks and accelerates the p75NTR retrograde axonal transport machinery. *PLoS Pathog.* **10**, e1004348 (2014). DOI: [10.1371/journal.ppat.1004348](https://doi.org/10.1371/journal.ppat.1004348).
- [10] Thoulouze, M.-I. et al. The neural cell adhesion molecule is a receptor for rabies virus. *J. Virol.* **72**, 7181–7190 (1998). DOI: [10.1128/JVI.72.9.7181-7190.1998](https://doi.org/10.1128/JVI.72.9.7181-7190.1998).
- [11] Lentz, T. L. et al. Is the acetylcholine receptor a rabies virus receptor? *Science* **215**, 182–184 (1982). DOI: [10.1126/science.7053569](https://doi.org/10.1126/science.7053569).
- [12] Willoughby, R. E. et al. Survival after treatment of rabies with induction of a coma. *N. Engl. J. Med.* **352**, 2508–2514 (2005). DOI: [10.1056/NEJMoa050382](https://doi.org/10.1056/NEJMoa050382).
- [13] Jackson, A. C. Therapy of human rabies. In *Rabies: Scientific Basis of the Disease and Its Management* (eds. Jackson, A. C. & Wunner, W. H.) 575–589 (Academic Press, London, 2013). DOI: [10.1016/B978-0-12-396547-9.00016-X](https://doi.org/10.1016/B978-0-12-396547-9.00016-X).
- [14] Jackson, A. C. Current and future approaches to the therapy of human rabies. *Antiviral Res.* **99**, 61–67 (2013). DOI: [10.1016/j.antiviral.2013.01.003](https://doi.org/10.1016/j.antiviral.2013.01.003).
- [15] Jackson, A. C. Failure of the Milwaukee Protocol for rabies. *Clin. Infect. Dis.* **57**, 1083–1085 (2013). DOI: [10.1093/cid/cit448](https://doi.org/10.1093/cid/cit448).
- [16] Jackson, A. C. Human rabies: neuropathogenesis, clinical management, and therapeutic horizons. *Vaccine* **42**, 3420–3428 (2024). DOI: [10.1016/j.vaccine.2024.03.045](https://doi.org/10.1016/j.vaccine.2024.03.045).
- [17] Jackson, A. C. Demise of the Milwaukee Protocol for rabies. *Clin. Infect. Dis.* **81**, e229–e232 (2025). DOI: [10.1093/cid/ciaf157](https://doi.org/10.1093/cid/ciaf157).
- [18] Brzozka, K., Finke, S. & Conzelmann, K.-K. Inhibition of interferon signaling by rabies virus phosphoprotein P: activation-dependent binding to STAT1 and STAT2. *J. Virol.* **80**, 2675–2683 (2006). DOI: [10.1128/JVI.80.6.2675-2683.2006](https://doi.org/10.1128/JVI.80.6.2675-2683.2006).
- [19] Vidy, A., Chelbi-Alix, M. & Blondel, D. Rabies virus phosphoprotein forms homo-oligomers and inhibits interferon signal transduction pathways. *J. Virol.* **79**, 14411–14420 (2005). DOI: [10.1128/JVI.79.22.14411-14420.2005](https://doi.org/10.1128/JVI.79.22.14411-14420.2005).
- [20] Nikolic, J. et al. Negri bodies are viral factories with properties of liquid organelles. *Nat. Commun.* **8**, 58 (2017). DOI: [10.1038/s41467-017-00102-9](https://doi.org/10.1038/s41467-017-00102-9).
- [21] Abbott, N. J. et al. Structure and function of the blood-brain barrier. *Neurobiol. Dis.* **37**, 13–25 (2010). DOI: [10.1016/j.nbd.2009.07.030](https://doi.org/10.1016/j.nbd.2009.07.030).
- [22] Pardridge, W. M. The blood-brain barrier: bottleneck in brain drug development. *NeuroRx* **2**, 3–14 (2005). DOI: [10.1602/neuroRx.2.1.3](https://doi.org/10.1602/neuroRx.2.1.3).
- [23] Sykova, E. & Nicholson, C. Diffusion in brain extracellular space. *Physiol. Rev.* **88**, 1277–1340 (2008). DOI: [10.1152/physrev.00027.2007](https://doi.org/10.1152/physrev.00027.2007).
- [24] Thorne, R. G. & Nicholson, C. In vivo diffusion analysis with quantum dots and dextrans predicts the width of brain extracellular space. *Proc. Natl. Acad. Sci. USA* **103**, 5567–5572 (2006). DOI: [10.1073/pnas.0509425103](https://doi.org/10.1073/pnas.0509425103).
- [25] Bobo, R. H. et al. Convection-enhanced delivery of macromolecules in the brain. *Proc. Natl. Acad. Sci. USA* **91**, 2076–2080 (1994). DOI: [10.1073/pnas.91.6.2076](https://doi.org/10.1073/pnas.91.6.2076).
- [26] Morrison, P. F. et al. High-flow microinfusion: tissue penetration and pharmacodynamics. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **266**, R292–R305 (1994). DOI: [10.1152/ajpregu.1994.266.1.R292](https://doi.org/10.1152/ajpregu.1994.266.1.R292).
- [27] Lonser, R. R., Sarntinoranont, M., Morrison, P. F. & Oldfield, E. H. Convection-enhanced delivery to the central nervous system. *J. Neurosurg.* **122**, 697–706 (2015). DOI: [10.3171/2014.10.JNS14229](https://doi.org/10.3171/2014.10.JNS14229).
- [28] Lonser, R. R. et al. Real-time image-guided direct convective perfusion of intrinsic brainstem lesions: technical note. *J. Neurosurg.* **107**, 190–197 (2007). DOI: [10.3171/JNS-07/07/0190](https://doi.org/10.3171/JNS-07/07/0190).

- [29] Krauze, M. T. et al. Reflux-free cannula for convection-enhanced high-speed delivery of therapeutic agents. *J. Neurosurg.* **103**, 923–929 (2005). DOI: [10.3171/jns.2005.103.5.0923](https://doi.org/10.3171/jns.2005.103.5.0923).
- [30] Sillay, K. A. et al. Convection-enhanced delivery of therapeutic agents to the brain: cannula design and development. *J. Neurosurg.* **120**, 1475–1483 (2014). DOI: [10.3171/2014.2.JNS1388](https://doi.org/10.3171/2014.2.JNS1388).
- [31] Warren, T. K. et al. Protection against filovirus diseases by a novel broad-spectrum nucleoside analogue BCX4430. *Nature* **508**, 402–405 (2014). DOI: [10.1038/nature13027](https://doi.org/10.1038/nature13027).
- [32] Zhao, J. et al. Lipid droplets are beneficial for rabies virus replication by facilitating viral budding. *J. Virol.* **96**, e01473-21 (2022). DOI: [10.1128/jvi.01473-21](https://doi.org/10.1128/jvi.01473-21).
- [33] Scott, C. A., Rossiter, J. P. & Jackson, A. C. Degeneration of neuronal processes in rabies virus encephalitis. *Front. Cell. Infect. Microbiol.* **10**, 568461 (2020). DOI: [10.3389/fcimb.2020.568461](https://doi.org/10.3389/fcimb.2020.568461).
- [34] Zhu, F.-C. et al. Efficacy and safety of the recombinant human monoclonal anti-rabies antibody cocktail SYN023 (zamorvimab and mazorelvimab) in post-exposure prophylaxis: integrated analysis of Phase 2b and randomized double-blind Phase 3 trials. *Emerg. Microbes Infect.* **13**, 2314567 (2024). DOI: [10.1080/22221751.2024.2314567](https://doi.org/10.1080/22221751.2024.2314567).
- [35] Terryn, S. et al. Protective effect of different anti-rabies virus VHH constructs against rabies disease in mice. *PLoS ONE* **9**, e109367 (2014). DOI: [10.1371/journal.pone.0109367](https://doi.org/10.1371/journal.pone.0109367).
- [36] Terryn, S. et al. Post-exposure treatment with anti-rabies VHH and vaccine significantly improves protection of mice from lethal rabies infection. *PLoS Negl. Trop. Dis.* **10**, e0004902 (2016). DOI: [10.1371/journal.pntd.0004902](https://doi.org/10.1371/journal.pntd.0004902).
- [37] Taylor, R. et al. BCX4430: a broad-spectrum antiviral adenosine nucleoside analog under development for the treatment of Ebola virus disease. *J. Infect. Public Health* **9**, 220–226 (2016). DOI: [10.1016/j.jiph.2016.04.002](https://doi.org/10.1016/j.jiph.2016.04.002).
- [38] Ren, Z. et al. A SynB1-conjugated antibody cocktail crosses the blood-brain barrier to produce a therapeutic effect on rabies. *Proc. Natl. Acad. Sci. USA* **122**, e2516465122 (2025). DOI: [10.1073/pnas.2516465122](https://doi.org/10.1073/pnas.2516465122).
- [39] Yu, Y. J. et al. Boosting brain uptake of a therapeutic antibody by reducing its affinity for a transcytosis target. *Sci. Transl. Med.* **3**, 84ra44 (2011). DOI: [10.1126/scitranslmed.3002230](https://doi.org/10.1126/scitranslmed.3002230).
- [40] Pardridge, W. M. Blood-brain barrier delivery of protein and antibody therapeutics. *Neurochem. Int.* **136**, 104720 (2020). DOI: [10.1016/j.neuint.2020.104720](https://doi.org/10.1016/j.neuint.2020.104720).
- [41] Ugolini, G. Rabies virus as a transneuronal tracer of neuronal connections. *Adv. Virus Res.* **79**, 165–202 (2011). DOI: [10.1016/B978-0-12-387040-7.00010-X](https://doi.org/10.1016/B978-0-12-387040-7.00010-X).
- [42] Jackson, A. C. & Park, H. Apoptotic cell death in experimental rabies in suckling mice. *Acta Neuropathol.* **95**, 159–164 (1998). DOI: [10.1007/s004010050781](https://doi.org/10.1007/s004010050781).
- [43] Huang, E. J. & Reichardt, L. F. Neurotrophins: roles in neuronal development and function. *Annu. Rev. Neurosci.* **24**, 677–736 (2001). DOI: [10.1146/annurev.neuro.24.1.677](https://doi.org/10.1146/annurev.neuro.24.1.677).
- [44] Park, H. & Poo, M.-M. Neurotrophin regulation of neural circuit development and function. *Nat. Rev. Neurosci.* **14**, 7–23 (2013). DOI: [10.1038/nrn3379](https://doi.org/10.1038/nrn3379).
- [45] Varendi, K., Kumar, A., Harma, M. & Andressoo, J.-O. microRNA-1 and microRNA-10b regulate brain-derived neurotrophic factor expression in mouse neurons. *J. Neurochem.* **138**, 429–440 (2016). DOI: [10.1111/jnc.13650](https://doi.org/10.1111/jnc.13650).
- [46] Lubin, F. D., Roth, T. L. & Sweatt, J. D. Epigenetic regulation of BDNF gene transcription in the consolidation of fear memory. *J. Neurosci.* **28**, 10576–10586 (2008). DOI: [10.1523/JNEUROSCI.1786-08.2008](https://doi.org/10.1523/JNEUROSCI.1786-08.2008).
- [47] Piccinotti, S. & Whelan, S. P. J. Rabies virus enters host cells by macropinocytosis and dynamin-dependent endocytosis. *J. Virol.* **90**, 9904–9917 (2016). DOI: [10.1128/JVI.01278-16](https://doi.org/10.1128/JVI.01278-16).
- [48] US Food and Drug Administration. *Expanded Access to Investigational Drugs for Treatment Use: Questions and Answers (Guidance for Industry, Revision 2)*. (FDA CDER/CBER, Silver Spring, MD, 2026). [FDA Guidance](https://www.fda.gov/oc/expanded-access-to-investigational-drugs-for-treatment-use).
- [49] US Food and Drug Administration. *Form FDA 3926: Individual Patient Expanded Access Applications (Guidance for Industry)*. (FDA CDER/CBER, Silver Spring, MD, 2017). [FDA Form 3926 Guidance](https://www.fda.gov/oc/expanded-access-to-investigational-drugs-for-treatment-use).
- [50] Dietzschold, B. et al. Characterization of an antigenic determinant of the glycoprotein of rabies virus associated with pathogenicity. *Proc. Natl. Acad. Sci. USA* **80**, 70–74 (1983). DOI: [10.1073/pnas.80.1.70](https://doi.org/10.1073/pnas.80.1.70).
- [51] Kuzmina, N. A. et al. Conservation of binding epitopes for monoclonal antibodies on the rabies virus glycoprotein. *J. Antivir. Antiretrovir.* **5**, 37–43 (2013). DOI: [10.4172/jaa.1000061](https://doi.org/10.4172/jaa.1000061).
- [52] Scott, C. A. et al. Structural abnormalities in neurons are sufficient to explain the clinical disease and fatal outcome of experimental rabies in yellow fluorescent protein-expressing transgenic mice. *J. Virol.* **82**, 513–521 (2008). DOI: [10.1128/JVI.01677-07](https://doi.org/10.1128/JVI.01677-07).
- [53] Cao, L. et al. Structural basis for broad neutralization of rabies virus by an antibody cocktail SYN023. *Emerg. Microbes Infect.* **14**, 2547724 (2025). DOI: [10.1080/22221751.2025.2547724](https://doi.org/10.1080/22221751.2025.2547724).
- [54] Brinker, T. et al. A new look at cerebrospinal fluid circulation. *Fluids Barriers CNS* **11**, 10 (2014). DOI: [10.1186/2045-8118-11-10](https://doi.org/10.1186/2045-8118-11-10).